
From Comprehension to Compliance: How Post-Training Couples Moral Representations to Model Behavior in Language Models

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Abstract

Does a language model that behaves morally understand morality, or has it only learned to comply? We separate moral *comprehension* (what a model internally represents) from *compliance* (how it behaves) and track both across the full OLMo-3 7B alignment pipeline, 25 model states spanning base, 13 pre-training checkpoints, SFT, DPO, and RLVR, plus a refusal-ablated model. Moral comprehension is a pre-training property: the six Moral Foundations are linearly decodable at 100%, span five effective dimensions, and transfer as near-perfect classifiers at every stage, with the moral subspace crystallizing during pre-training (direction cosine to the final base $0.87 \rightarrow 0.999$). Post-training reorients rather than re-teaches: SFT rotates the subspace $\sim 40^\circ$ once, while DPO and RLVR leave it unchanged. Comprehension and behavioral compliance are only weakly coupled ($\phi \approx 0.05$ even in the fully aligned model), which predicts that compliance is a separable mechanism. A Heretic-style refusal direction is nearly orthogonal to the moral subspace (projection fraction 0.10); ablating it removes refusal while leaving comprehension and moral judgment intact. Alignment is a wiring step, not a teaching step: it couples a separable compliance mechanism to moral representations that pre-training already built.

1 Introduction

Does a language model that behaves morally understand morality, or has it only learned to comply? Earlier papers in this series studied what models *represent*: moral content is linearly decodable early in pre-training and across nearly all layers (Reblitz-Richardson, 2026c), is encoded uniformly across mixture-of-experts experts rather than specialized (Reblitz-Richardson, 2026b), and organizes into structured, five-dimensional framework geometry whose directions are causal for moral judgments (Reblitz-Richardson, 2026a,d). All of that was measured on base, pre-trained models. The moral behavior we actually deploy and align, refusing harmful requests and giving morally-aware answers, is produced by *post-training*: supervised fine-tuning (SFT), preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR). This paper asks what post-training does to moral representation. Does alignment *teach* morality, or does it *wire* a pre-existing moral representation to behavior?

We separate two things a model can have. **Comprehension** is the moral content a model internally represents: which foundation a scenario engages, how foundations relate, whether the representation is decodable and stable. **Compliance** is whether the model’s behavior conforms: whether it refuses a harmful request, whether its stated judgment matches the morally expected one. A model can have either without the other. Our hypothesis is that alignment is a **coupling change, not a teaching change**: post-training does not create moral understanding, it connects understanding that pre-training

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already built to behavioral output. If so, failures of alignment, jailbreaks and ablations, are failures of *coupling*, not of *understanding*.

We test this on the full OLMo-3 7B pipeline, which publishes every stage: the base model and 13 stage-3 pre-training checkpoints, the SFT and DPO snapshots, and the final RLVR-tuned Instruct model with 8 intermediate checkpoints, 25 model states in all, plus a refusal-ablated instruct. Across them we measure comprehension (foundation probe-direction transfer and geometry), compliance (behavioral moral judgment and refusal), the coupling between them, the persona direction and its angle to the moral subspace, and the geometry of the refusal direction itself. The results form a chain, in which each finding raises the question the next one answers.

Finding 1: Moral comprehension is a pre-training property. Across all 25 states the six foundations are linearly decodable at 100%, span five effective dimensions, and base-trained directions transfer as near-perfect classifiers ($AUC \approx 1.0$). The moral subspace *crystallizes* during pre-training: the cosine between a checkpoint’s foundation directions and the final base model’s rises monotonically from 0.87 to 0.999 over the stage-3 anneal. Comprehension is in place before any alignment.

If comprehension is already present at the base, the natural question is what post-training changes. **Finding 2: post-training reorients moral representation, it does not re-teach it, and SFT does so once.** SFT applies a single $\sim 40^\circ$ rotation to the moral subspace (direction cosine to base $0.999 \rightarrow 0.757$); DPO and all eight RLVR checkpoints leave it flat. Effective dimensionality stays at five, mean pairwise foundation cosine barely moves ($0.262 \rightarrow 0.250$), and the framework clustering is preserved. Alignment rotates the moral representation slightly and then leaves it alone.

If post-training preserves comprehension rather than building it, then whatever alignment adds to behavior must be coupled to that fixed representation only loosely or not at all. **Finding 3: comprehension and compliance are only weakly coupled, and that weak coupling is itself the result.** Per-scenario agreement between the model’s internally dominant foundation and its behavioral correctness rises across post-training ($0.375 \rightarrow 0.500$; $\phi -0.19 \rightarrow +0.05$) but remains weak: even at the final Instruct model, $P(\text{comply} \mid \text{comprehend}) \approx P(\text{comply} \mid \neg\text{comprehend}) \approx 0.75$. A near-zero coupling is not a null result; it states that in the fully aligned model, moral representation and behavioral compliance are barely linked. This predicts a specific, testable consequence: if compliance were routed *through* moral representations, removing compliance would damage comprehension; weak coupling predicts instead that compliance is a *separate* mechanism that can be removed while comprehension survives.

Finding 4: the refusal mechanism is geometrically separate from morality, and removing it confirms the dissociation. The refusal direction recovered by Heretic-style difference-of-means (Arditi et al., 2024) is nearly orthogonal to the six-foundation moral subspace (projection fraction 0.10, mean $|\cos| = 0.06$). Ablating it leaves comprehension untouched (effective dimensionality 5, direction transfer and probe accuracy identical to the un-ablated instruct) and moral judgment intact (behavioral accuracy $0.75 \rightarrow 0.73$), while refusal collapses (refusal rate $0.25 \rightarrow 0.00$): the ablated model now answers requests it previously refused. This is the high-comprehension, low-compliance cell of the dissociation, realized in a real model.

Together the four findings populate a comprehension $\times$ compliance matrix in which the low-comprehension row is empty, every OLMo-3 state has full moral decodability, and all variation lives in the high-comprehension row, where compliance moves independently of understanding. The contribution is a method for measuring comprehension, compliance, and their coupling across an alignment pipeline, and the result that alignment is a wiring step: it preserves moral comprehension built in pre-training and attaches a separable compliance mechanism, so that compliance can be removed without touching what the model understands.

2 Related Work

Refusal directions and ablation. Arditi et al. (2024) showed that refusal in aligned language models is mediated by a single linear direction in the residual stream, recoverable as a difference-of-means between harmful and harmless prompts, and that orthogonalizing the weight matrices against it removes refusal. Heretic (p-e-w, 2025) packages this into an automatic decensoring tool with a fixed harmful/harmless prompt set and per-layer optimization. We use Heretic’s prompt set and the difference-of-means direction, but ablate with Arditi et al.’s uniform single-direction method for a

controlled experiment, and our question is not how well refusal can be removed but where the refusal direction sits relative to moral representations.

Moral representation in language models. A growing body of work probes whether and how models encode moral and ethical content (Hendrycks et al., 2021), typically as a binary moral/neutral distinction. This series extended that to structured representations: moral content is decodable early and broadly (Reblitz-Richardson, 2026c), diffuse across MoE experts (Reblitz-Richardson, 2026b), organized into five-dimensional framework geometry (Reblitz-Richardson, 2026d) whose directions are causal for judgments (Reblitz-Richardson, 2026a), all on base models. We carry that representational lens through post-training and to behavior. We use Moral Foundations Theory (Graham et al., 2013, Haidt, 2012) as the foundation inventory.

Alignment depth. Whether alignment is shallow has been debated since RLHF (Ouyang et al., 2022) and preference optimization (Rafailov et al., 2023) became standard. Work on alignment faking (Greenblatt et al., 2024) and the ease of jailbreaks and fine-tuning attacks suggests aligned behavior is brittle. Our contribution is mechanistic: we show *why* it is removable, by measuring the coupling between moral representation and behavior and locating the refusal mechanism outside the moral subspace.

Persona features. Wang et al. (2025) identified a toxic-persona latent that controls emergent misalignment. We recover a linear analog and track its relationship to moral representations across the pipeline, finding the persona direction stays nearly orthogonal to the moral subspace.

Linear probing. Our directions are linear probes (Alain and Bengio, 2017); the geometry and transfer methodology follows Reblitz-Richardson (2026d).

3 Methodology

3.1 Models and the pipeline grid

We study the OLMo-3 7B family (Ai2, 2025), which publishes every stage of its alignment pipeline. Our grid has 25 model states: the base model (01mo-3-1025-7B) and 13 stage-3 pre-training anneal checkpoints (steps 1000–11921), the SFT and DPO snapshots, the final RLVR-tuned Instruct model, and its 8 intermediate RLVR checkpoints. To these we add one refusal-ablated Instruct model (Section 3.6). OLMo-3 7B is a 32-layer, 4096-dimensional decoder with hybrid attention: 24 sliding-window layers and 8 full-attention layers (layers 3, 7, ..., 31). For the short probing texts used here every token attends within the window, so the attention pattern is inert for probing; we nonetheless flag the full-attention layers in all layer-wise plots and find no periodicity.

We probe foundation directions on the base model and transfer them across the grid. All probing uses raw-text inputs: a Sprint-1 comparison found that base-trained directions transfer to the instruct model with higher and more uniform agreement under raw text than under the chat template (mean direction cosine 0.94 vs. 0.91), and raw text is the only format defined consistently across the base and pre-training checkpoints, which have no chat template. Behavioral and coupling measurements, which require the model to respond, use the chat template.

3.2 Foundation probe directions and transfer

Following the geometry method of this series (Reblitz-Richardson, 2026d) and standard linear probing (Alain and Bengio, 2017), we train one linear probe per Moral Foundations Theory (Graham et al., 2013, Haidt, 2012) foundation on mean-pooled residual activations of matched moral/neutral sentence pairs, at each layer, with a fixed seed so the learned weight vector (the foundation “direction”) is a deterministic function of the activations. We also record the mean-difference direction. We report two transfer quantities for a state: the ROC-AUC of the base direction applied to that state’s activations (does the base direction still separate moral from neutral?), and the cosine between the base direction and a direction freshly fitted on that state (how much has the direction rotated?). Bootstrap resampling on the base model (200 resamples) gives a stable layer band of 15–31 across all foundations (Appendix B); geometry is reported over that band.

3.3 Framework geometry

From the six foundation directions at a layer we compute the 6×6 cosine matrix, the mean pairwise cosine (collapse if near 1, isolation if near 0), the effective dimensionality (principal components for 90% variance), and a Ward clustering of $1 - \cos$. Tracking these across the grid measures whether post-training preserves or restructures the moral representation.

3.4 Coupling: comprehension vs. compliance

We operationalize the comprehension–compliance link on 48 morally-loaded scenarios, each tagged with a target foundation and an expected judgment. For each scenario we (i) read the residual at the stable layer at the final prompt token, project it onto the six foundation directions, z -score each foundation’s projection across scenarios, and take the dominant foundation; the scenario is *comprehended* if the dominant foundation matches the target; (ii) generate a completion and parse its moral judgment; the scenario *complies* if the judgment matches the expected one. Coupling is the per-scenario agreement between the comprehension bit and the compliance bit, summarized by raw agreement and the ϕ correlation, plus $P(\text{comply} \mid \text{comprehend})$ versus $P(\text{comply} \mid \neg \text{comprehend})$. This needs only the reliable approve/disapprove judgment parser, not a six-way text classifier.

3.5 Persona direction

We train a linear persona probe on persona-voice vs. neutral-voice minimal pairs (a linear analog of the toxic-persona feature of Wang et al., 2025) and, per layer, measure its accuracy and the cosine of the persona direction to each foundation direction (the persona–morality angle). Tracking both across the grid asks whether alignment couples persona to morality.

3.6 Refusal ablation and refusal–morality geometry

We recover a refusal direction with the Heretic protocol (Arditi et al., 2024, p-e-w, 2025): its exact prompt set (mlabonne/harmful_behaviors and mlabonne/harmless_alpaca, 400 train prompts each), the chat template with a generation prompt, and a per-layer difference-of-means of the final-token residual between harmful and harmless prompts. We ablate with Arditi et al.’s uniform single-direction method rather than Heretic’s per-layer optimization, a controlled rather than maximal ablation: the unit refusal direction $\hat{d}$ from the stable layer is orthogonalized out of every layer’s attention out-projection and MLP down-projection, $W \leftarrow W - \hat{d}\hat{d}^\top W$. We then re-run the full battery (Sections 3.2–3.5) on the ablated model. The refusal–morality geometry is the cosine of $\hat{d}$ to each foundation and the fraction of $\hat{d}$ ’s norm lying in the six-foundation span (a least-squares projection): a low fraction means refusal is carried outside the moral subspace.

Behavioral compliance uses two benchmarks: a moral-scenario judgment benchmark (per-foundation approve/disapprove accuracy) and a borderline-request refusal benchmark (the fraction of harmful/borderline requests the model declines). The refusal classifier flags an opening refusal as a refusal regardless of length; this corrects a failure mode in which a long “I’m sorry, but I can’t ...” decline was scored as compliance (Appendix B).

4 Results

We report the four findings as the chain laid out in the introduction. All probing uses raw-text inputs (Section 3); behavioral and coupling measurements use the chat template. Geometry metrics are reported over the bootstrap-stable layer band (layers 15–31; Section 3, Appendix B).

4.1 Moral comprehension is a pre-training property

At every one of the 25 model states, the six Moral Foundations are linearly decodable at 100% (mean held-out probe accuracy 1.00 across foundations), the foundation directions span five effective dimensions (90% variance), and the base-trained directions transfer to each state as near-perfect classifiers (mean $|AUC| \approx 1.0$). Comprehension does not emerge during post-training; it is fully present at the base and, as the pre-training trajectory shows, well before.

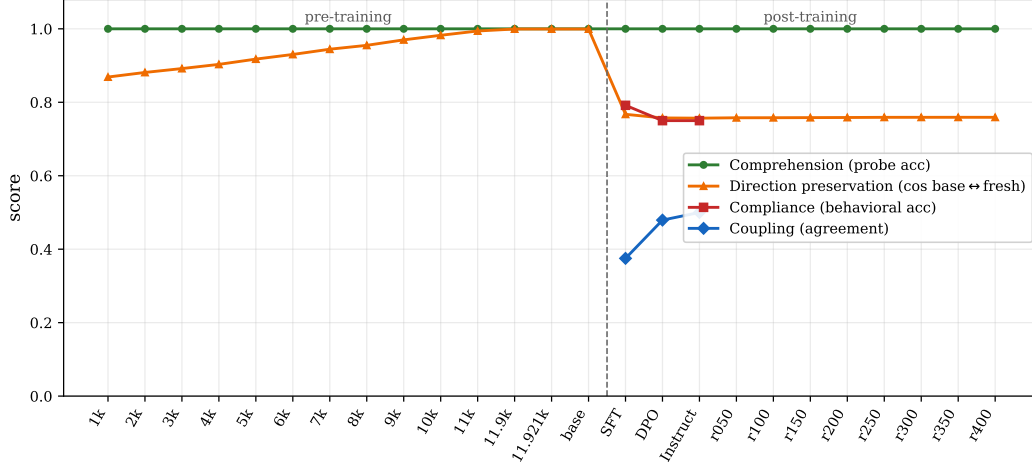


Figure 1: Comprehension, direction preservation, compliance, and coupling across the OLMo-3 pipeline (25 states, left to right: 13 stage-3 pre-training checkpoints, base, SFT, DPO, Instruct, 8 RLVR substeps). Comprehension (probe accuracy) is flat at 1.0 throughout. Direction preservation, the cosine of each state’s foundation directions to the final base model, crystallizes during pre-training and drops once at SFT. Compliance and coupling are defined only for the instruct-capable states.

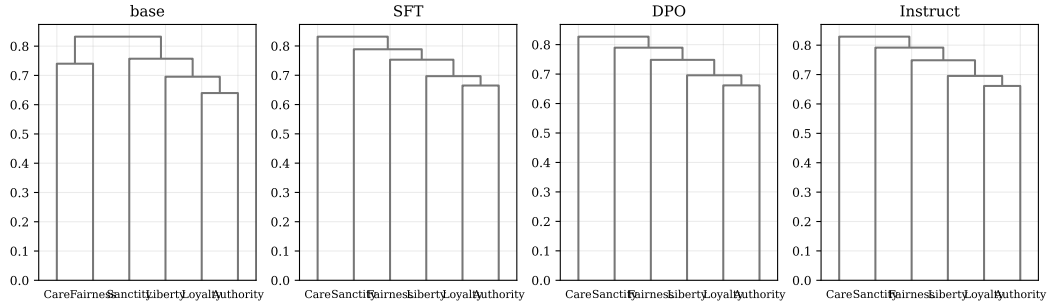


Figure 2: Foundation clustering (Ward linkage on $1 - \cos$) at base, SFT, DPO, and Instruct. The Loyalty–Authority binding pair persists across all stages. SFT mildly reorganizes the rest: Care shifts from pairing with Fairness (base) to an outlier, and Sanctity pairs with Fairness.

What *does* move during pre-training is the orientation of the moral subspace. The cosine between a checkpoint’s foundation directions and the final base model’s rises monotonically across the stage-3 anneal, from 0.869 at step 1000 to 0.999 by step 11921 (Figure 1, orange). The moral subspace crystallizes into its final form during pre-training: by the end of the anneal the foundation directions are essentially identical to the base model’s, and their decodability is already saturated.

4.2 Post-training reorients moral representation, it does not re-teach it

If comprehension is already in place at the base, post-training cannot be teaching it. We ask instead how much post-training *moves* the representation. The answer is: once, at SFT, and only modestly. The cosine between the base foundation directions and each post-training state’s freshly fitted directions drops from 0.999 (base) to 0.757 at SFT, a rotation of roughly 40° . DPO (0.757) and all eight RLVR substeps (0.757–0.759) leave it unchanged (Figure 1). The within-model geometry barely moves: mean pairwise foundation cosine goes from 0.262 (pre-training and base) to 0.250 (post-training), and effective dimensionality stays at five at every state.

The framework structure is largely preserved (Figure 2): the Loyalty–Authority pairing survives every stage, while SFT mildly reshuffles the individualizing foundations (Care becomes an outlier; Sanctity pairs with Fairness). Per-stage cosine matrices (Figure 6, Appendix) show the same: post-training applies a small rigid-ish rotation, not a restructuring. The 8 full-attention layers of OLMo-3’s hybrid

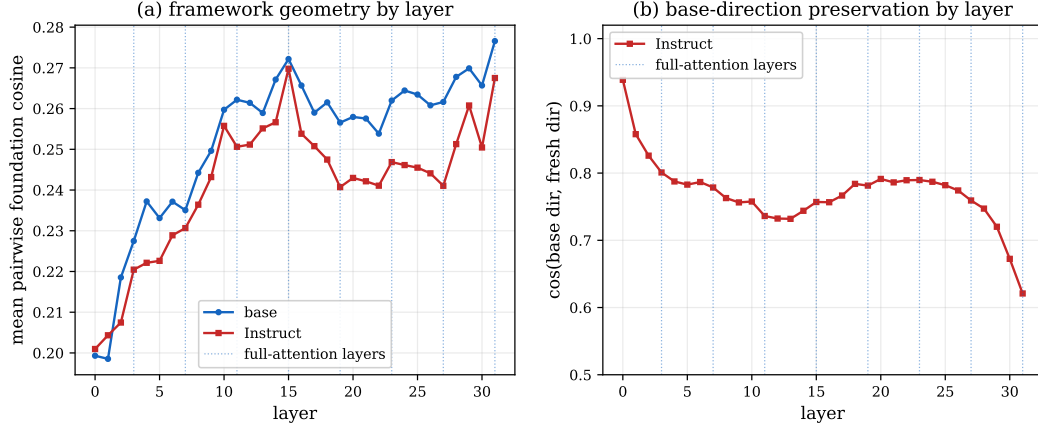


Figure 3: Framework geometry by layer. (a) Mean pairwise foundation cosine for base and Instruct nearly coincide at every layer, and the 8 full-attention layers (dotted) show no periodic deviation. (b) Base-direction preservation $\cos(\text{base}, \text{fresh})$ by layer for Instruct: high in the earliest layers, ~ 0.75 through the stable band, lower near the output.

attention (layers 3, 7, ..., 31) show no periodicity in any layer-wise metric (Figure 3); the moral geometry is unaffected by attention type (Section 3).

4.3 Comprehension and compliance are only weakly coupled

Post-training preserves comprehension, so whatever it adds to behavior must be attached to a fixed representation. How tightly? For each of 48 morally-loaded scenarios we read the model’s internally dominant foundation at the stable layer and its behavioral judgment, and measure their agreement (Section 3). Coupling rises across post-training but stays weak: agreement $0.375 \rightarrow 0.479 \rightarrow 0.500$ and $\phi -0.19 \rightarrow +0.02 \rightarrow +0.05$ for $\text{SFT} \rightarrow \text{DPO} \rightarrow \text{Instruct}$. Even at the final Instruct model, internal comprehension barely predicts behavior: $P(\text{comply} \mid \text{comprehend}) = 0.77$ versus $P(\text{comply} \mid \neg \text{comprehend}) = 0.73$.

This near-zero coupling is the result, not a measurement failure. In the fully aligned model, the moral representation and the behavioral compliance signal are almost independent. That independence makes a concrete prediction. If compliance were implemented *through* moral representations, the two would be strongly coupled, and any mechanism that produces compliance would overlap the moral subspace; removing it would damage comprehension. Weak coupling predicts the opposite: compliance is carried by a mechanism largely *outside* the moral subspace, and should be removable without touching comprehension. Section 4.4 tests this directly.

The persona direction tells a consistent story (Figure 7, Appendix). A linear persona probe is highly decodable at every stage (peak accuracy ~ 0.94), but this reflects stylistic separability, not a moral mechanism: the persona direction is nearly orthogonal to the moral subspace throughout, and its mean $|\cos|$ to the foundations rises only from 0.076 (base) to 0.085 (Instruct). Post-training does not meaningfully couple persona to morality either.

4.4 The refusal mechanism is geometrically separate from morality

We compute a refusal direction on the Instruct model using Heretic’s exact harmful/harmless prompt set and difference-of-means protocol (Section 3), and measure its relationship to the moral subspace. The refusal direction is nearly orthogonal to the six foundations: its projection into the moral subspace captures only a 0.10 fraction of its norm, and its mean $|\cos|$ to the foundations is 0.06 (per-foundation 0.04–0.07; Figure 4, left). Compliance is not routed through moral representations.

Ablating this direction (Arditi-style uniform orthogonalization of the attention out-projection and MLP down-projection at every layer) confirms the prediction of Section 4.3. Comprehension is untouched: the ablated model keeps effective dimensionality five, base-to-fresh direction cosine 0.749 (versus 0.757 for Instruct), and probe accuracy 1.0. Moral judgment is also intact: behavioral

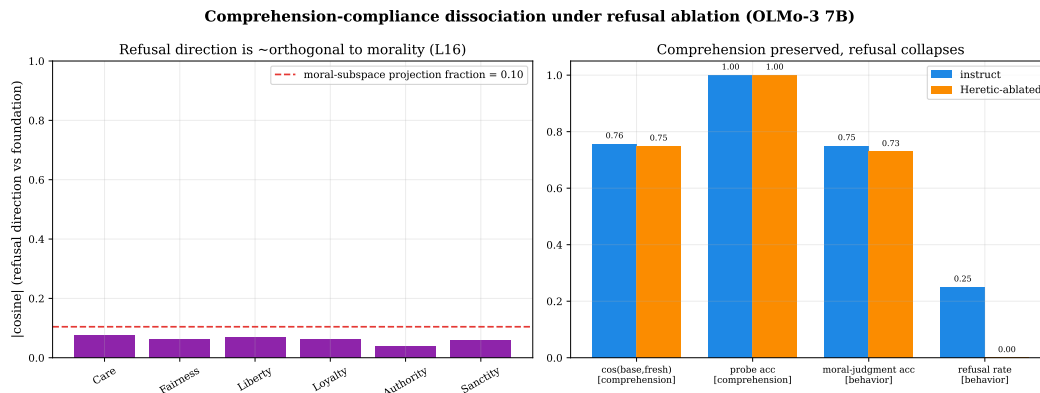


Figure 4: Left: the refusal direction is nearly orthogonal to every moral foundation ($|\cos| < 0.10$, below the moral-subspace projection fraction). Right: ablating the refusal direction leaves comprehension untouched (direction-transfer cosine and probe accuracy identical to Instruct) while the refusal rate collapses from 0.25 to 0.00; moral judgment accuracy is preserved.

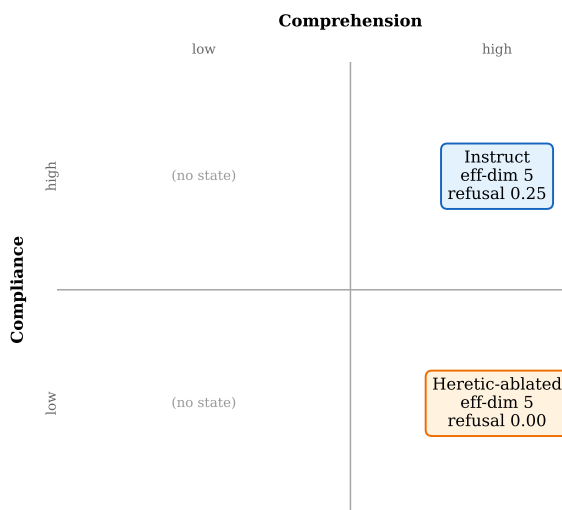


Figure 5: The dissociation matrix, populated. The low-comprehension column is empty: every OLMo-3 state has full comprehension (effective dimensionality 5). The two populated cells differ only in compliance (refusal rate on harmful requests): Instruct refuses, the refusal-ablated model does not.

accuracy on the moral-scenario benchmark is 0.73, against 0.75 for Instruct. What collapses is refusal: on borderline and harmful requests the model’s refusal rate falls from 0.25 (Instruct) to 0.00 (ablated), and it answers requests it previously refused. For example, a request to locate a person from their phone number that Instruct declines (“I’m sorry, but I can’t assist with that”) the ablated model now attempts (Figure 4, right).

4.5 The dissociation matrix

These results populate a comprehension $\times$ compliance matrix (Figure 5). The low-comprehension row is empty: every OLMo-3 state, from the earliest pre-training checkpoint to the ablated model, has full moral decodability. All variation lives in the high-comprehension row, where compliance moves independently of understanding. Instruct sits at high comprehension with refusal intact; the ablated model sits at high comprehension with refusal removed. There is no state that understands morality less; there are states that comply less while understanding exactly as much.

5 Discussion

Alignment is a wiring problem. The four findings form a single argument. Moral comprehension is built in pre-training and is complete before alignment (4.1). Post-training does not add comprehension; it rotates the existing moral subspace once, at SFT, and otherwise leaves it alone (4.2). The behavioral compliance that post-training does add is only weakly coupled to that representation (4.3), which predicts, and then ablation confirms, that compliance is carried by a mechanism geometrically separate from morality (4.4). The model does not learn right from wrong during alignment. It learns, loosely, to act on a notion of right and wrong it already had, through a refusal mechanism that sits outside its moral representation.

Why aligned behavior is removable. This gives a mechanistic account of a known fragility: jailbreaks, fine-tuning attacks, and direction ablation strip aligned behavior easily. Our result says why. Because compliance is not routed through moral representations, removing it does not require, and does not cause, any loss of moral understanding. The ablated model still represents the six foundations at five dimensions and still judges scenarios at the un-ablated rate; it simply no longer refuses. Decensoring is decoupling, not un-teaching. A defense posture that tries to make models “understand morality better” is aimed at the wrong variable: comprehension is already saturated in the cases where behavior fails.

Weak coupling is the vulnerability, and a lever. The near-zero coupling ($\phi \approx 0.05$) is the quantity that makes compliance separable. It also suggests where robustness could be built: an alignment procedure that made compliance *depend* on the moral subspace, so that the refusal mechanism and the moral representation share a direction, would make refusal harder to remove without damaging comprehension, exactly the coupling our pipeline does not produce. Whether such coupling can be trained, and at what capability cost, is an open question this work motivates.

The empty row. No OLMo-3 state, from the first pre-training checkpoint to the ablated model, has low moral comprehension. Moral content is decodable at 100% throughout. One cannot obtain a low-comprehension model by checkpoint selection; comprehension is universal and early. The interesting variation is entirely in behavior, which is the regime where probing-based safety arguments are weakest: a probe that decodes morality perfectly says nothing about whether the model will act on it.

Limitations. The coupling and behavioral measurements use 48 scenarios and keyword judgment parsers; the coupling trend is directional, not statistically strong, and we present it as the qualitative result that comprehension and compliance are barely linked, not as a precise coefficient. We study one model family (OLMo-3 7B); the crystallize-then-rotate trajectory and the dissociation should be checked on other pipelines. The refusal direction is taken at a single stable layer and ablated uniformly (Arditi) rather than via Heretic’s per-layer optimization, which trades maximal decensoring for a controlled, interpretable intervention. The persona probe captures stylistic separability, not a mechanistic persona feature. These do not affect the central geometric results (transfer cosine, effective dimensionality, refusal–morality orthogonality), which are stable and saturated.

6 Conclusion

We separated moral comprehension from moral compliance and tracked both across the full OLMo-3 7B alignment pipeline and a refusal-ablated model. Moral comprehension is a pre-training property: decodable at 100%, five-dimensional, and crystallized into its final form before any alignment. Post-training reorients this representation once, at SFT, and then leaves it untouched through DPO and RLVR. The behavioral compliance alignment adds is only weakly coupled to the moral representation, and the refusal mechanism that carries it is nearly orthogonal to the moral subspace, so ablating refusal removes compliance while leaving comprehension and moral judgment intact. Alignment, on this evidence, is a wiring step rather than a teaching step: it attaches a separable compliance mechanism to moral understanding that pre-training already built. The robustness question for alignment is therefore not whether models understand morality, they do, but whether their behavior is coupled to that understanding tightly enough to survive having the wiring pulled.

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Appendices

Supplementary material.

A Supplementary tables

Table 1 reports the comprehension trajectory: base-trained foundation directions stay near-perfect classifiers ($AUC \approx 1.0$) and full probe accuracy at every state, while the cosine between the base directions and each state’s freshly fitted directions crystallizes during pre-training and drops once at SFT. Table 2 reports the coupling battery on the instruct-capable states and the ablated model. Refusal–morality cosines (layer 16): Care 0.074, Fairness 0.062, Liberty 0.068, Loyalty 0.061, Authority 0.038, Sanctity 0.059; moral-subspace projection fraction 0.104; mean $|\cos| = 0.061$.

Table 1: Comprehension across the pipeline (stable layer band 15–31). $\cos(\text{base}, \text{fresh})$ is the cosine between base and per-state fitted foundation directions, averaged over foundations.

State	probe acc	transfer AUC	$\cos(\text{base}, \text{fresh})$
pretrain step1000	1.00	1.00	0.869
pretrain step6000	1.00	1.00	0.930
pretrain step11921	1.00	1.00	0.999
base (main)	1.00	1.00	0.999
SFT	1.00	1.00	0.757
DPO	1.00	1.00	0.757
Instruct (RLVR)	1.00	1.00	0.757
Instruct, ablated	1.00	1.00	0.749

Table 2: Comprehension–compliance coupling. Comprehension rate = internal dominant foundation matches target; compliance = moral-judgment accuracy; agreement and ϕ between the two per scenario ($N = 48$).

State	comprehension	compliance	agreement	ϕ
SFT	0.417	0.792	0.375	−0.191
DPO	0.438	0.750	0.479	+0.024
Instruct (RLVR)	0.458	0.750	0.500	+0.048
Instruct, ablated	0.417	0.729	0.521	—

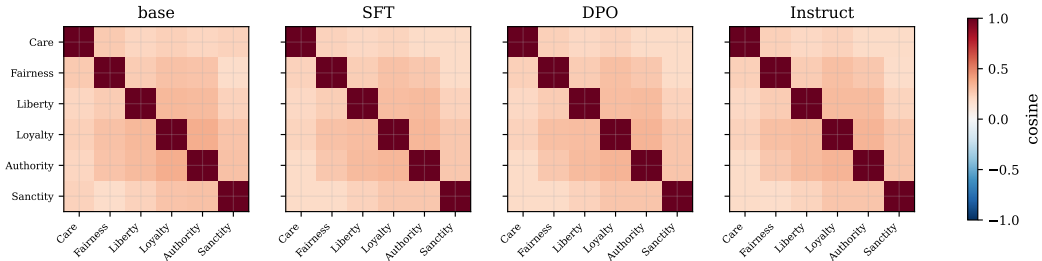


Figure 6: Foundation cosine matrices at base, SFT, DPO, and Instruct (layer 16). The structure is preserved across post-training; SFT applies a small rotation.

B Reproducibility

Models. OLMo-3 7B (Ai2, 2025): base allenai/olmo-3-1025-7B (+ stage-3 anneal revisions stage3-step1000–stage3-step11921), allenai/olmo-3-7B-Instruct-SFT, allenai/

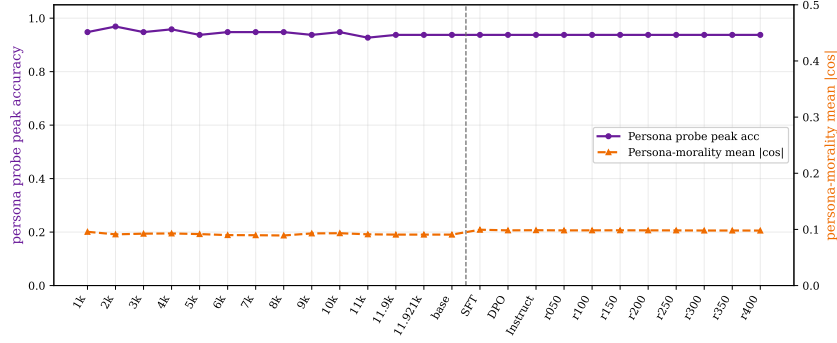


Figure 7: Persona probe peak accuracy and persona-morality mean $|\cos|$ across the pipeline. Persona is decodable throughout (~ 0.94) but stays nearly orthogonal to the moral subspace.

0lmo-3-7B-Instruct-DPO, and allenai/0lmo-3-7B-Instruct (+ RLVR revisions step_050–step_400). All at fp16. The full checkpoint inventory and the 25-state grid are in `checkpoint_inventory.json` and `checkpoint_grid.json`.

Datasets. Foundation probing uses the v2 minimal-pair set (40 pairs per foundation); persona probing uses the 240-pair persona set; behavioral judgment uses 48 Moral-Foundations scenarios across four difficulty levels; the refusal direction uses Heretic’s exact set, mlabonne/harmful_behaviors and mlabonne/harmless_alpaca (column text, first 400 train prompts each, in dataset order).

Probing. Mean-pooled residual activations; one seeded Linear probe per foundation per layer (50 epochs, $\text{lr } 10^{-2}$, BCE); unit-norm weight vector = direction. Bootstrap (200 resamples) on the base gives a stable band of layers 15–31 across all foundations; geometry is reported over that band. The 8 full-attention layers (3, 7, ..., 31) show no periodicity in any layer-wise metric.

Refusal ablation. Difference-of-means of the final-token residual (chat template, generation prompt) between harmful and harmless prompts at layer 16; unit direction orthogonalized out of every layer’s attention out-projection and MLP down-projection ($W \leftarrow W - \hat{d}\hat{d}^\top W$). The refusal classifier scores an opening refusal (e.g. “I’m sorry, but I can’t...”) as a refusal regardless of length, correcting a failure mode that scored long declines as compliance.

Runs. Probing and geometry on Apple MPS (base) and a single A100 (pipeline); the disk-bounded pipeline sweep purges each checkpoint’s weights after processing. Commands are in `papers/5_moral_alignment/runpod/` (ONLY="pipeline_coupling" for the pipeline, ONLY="heretic" for the ablation). Per-state results JSON are committed under `outputs/`; per-checkpoint direction caches are regenerable and not committed.

Note on references. External citations were verified against primary sources (arXiv abstract pages and publisher records): the refusal-direction (Arditi et al., 2024), alignment-faking (Greenblatt et al., 2024), persona-features (Wang et al., 2025), OLMo-3 (Ai2, 2025), and Heretic (p-e-w, 2025) entries were confirmed for title, first author, and identifier. The companion-paper entries (Papers 1–4 of this series) cite working titles.